

Estimation of Nonlinear Intra- and Inter-Channel Impairments in Optical Fiber Transmission

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Abstract We propose a neural network to estimate the relative contributions of intra- and inter-channel nonlinear impairments in fiber optic transmission from received-signal power spectral densities. Trained and tested on experimental data, the model achieves 1% RMSE across varying launch powers, symbol rates, and transmission distances. ©2026 The Author(s)

Introduction

Self-phase modulation (SPM), the dominant intra-channel nonlinearity, arises when the phase of a channel is modulated proportionally to its own power. In contrast, cross-phase modulation (XPM), the dominant inter-channel nonlinearity, occurs when the phase of a channel is modulated by the power of adjacent channels in wavelength-division multiplexing (WDM) systems.

Despite their importance in the network, SPM and XPM are not directly observable in practice. However, their measurement could unlock a range of capabilities in network optimization and modeling, enabling adaptive power optimization by balancing nonlinear distortions across channels [1], facilitating the integration of alien wavelengths by guiding their power settings to limit XPM-induced degradation on existing channels while maintaining their own performance [2], supporting the validation and calibration of physical-layer models where separating SPM and XPM is otherwise difficult [3], and improving performance prediction under evolving conditions such as new modulation formats, symbol rates [4], or new nonlinearity compensation strategies.

Several techniques have been proposed to monitor and characterize inter-channel nonlinear effects (XPM) [5–7], but existing approaches rely primarily on pilot-based techniques, which reduce spectral efficiency by allocating part of the transmission signal to non-data symbols.

In this paper, we propose a digital signal processing (DSP)-based approach to estimate the contributions of SPM and XPM, as well as the overall nonlinear interference (NLI), directly from the received signal. We develop a neural network that uses digitally recovered power spectral densities (PSDs) as input to estimate as outputs the relative contributions of nonlinear impairments accumulated during transmission. These outputs are reported as %SPM, %XPM, and %NLI.

Data Generation and Neural Network Design

To estimate the nonlinear impairment ratios, we

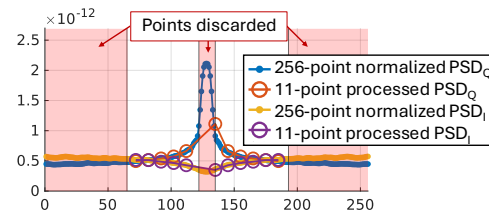


Fig. 1: Normalized PSD_Q/PSD_I spectra and their processed 11-point representations.

propose a neural network that takes as input a feature vector derived from the PSDs of the imaginary and real components of the signal distortion, denoted PSD_Q and PSD_I , respectively. The distortion is defined as the complex residual obtained by subtracting the ideal transmitted constellation points from the DSP-recovered symbols. The network outputs the estimated nonlinear impairment ratios; %SPM, %XPM, and %NLI.

The PSD_Q and PSD_I spectra (256 points each), are normalized by the distortion variance and smoothed using a 30-point moving average. As illustrated in Fig. 1, we discard 25% of the edge frequencies and 5% around zero frequency, as these regions provide limited information and may introduce overfitting. The remaining spectra are then downsampled with a 10% step, resulting in 11-point PSD_Q and PSD_I vectors. These are combined with the symbol rate, accumulated chromatic dispersion, and distortion variance to form a 25-element input feature vector. Finally, all features are normalized to the range [0, 1].

Our neural network consists of a single hidden layer with 15 neurons and hyperbolic tangent sigmoid activation functions. It is trained using the Bayesian regularization algorithm with mean squared error as the loss function, and its performance is evaluated using 5-fold cross-validation.

Training and testing data are obtained from point-to-point transmissions under various launch power profiles using the setup described in the next section, while the corresponding input features and ground-truth labels are derived from the experimental data acquisition procedure.

Experimental Setup

The experimental setup is illustrated in Fig. 2.

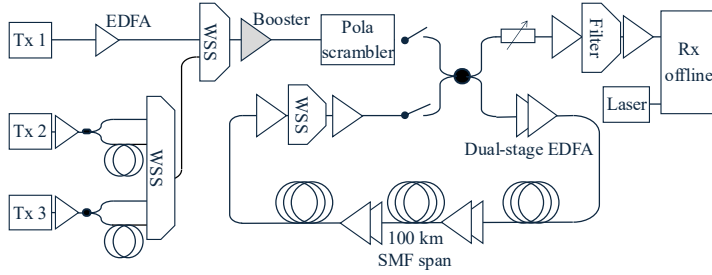


Fig. 2: Experimental setup. Tx: transmitter, WSS: wavelength selective switch, VOA: variable optical attenuator, Rx: receiver.

The transmission line consists of a loop of three spans of single-mode fiber (SMF), each 100 km long, with an attenuation of 0.21 dB/km, an effective area of 80 μm^2 , and chromatic dispersion of 16.7 ps/nm/km. Dual-stage erbium-doped fiber amplifiers (EDFAs) were used along the line, providing an output power per channel ranging from -7 to 10 dBm. Fifteen dual-polarization quadrature phase-shift keying (QPSK) channels were modulated at symbol rates of 32 GBd, 39 GBd, and 46 GBd, with 50 GHz channel spacing. The QPSK channels were shaped by a root-raised-cosine filter with a roll-off factor of 0.01, and the central channel, the channel under test (CUT), was set at 1545.4 nm. One offline transponder generated the CUT (Tx1), while two others (Tx2 and Tx3) with delay lines generated the fourteen adjacent channels.

At the coherent receiver, the DSP chain includes chromatic dispersion compensation using a finite impulse response (FIR) filter, followed by 25-tap constant-modulus algorithm (CMA)-based polarization demultiplexing, carrier frequency estimation (CFE), and carrier phase estimation (CPE) using a 201 taps Viterbi-Viterbi algorithm.

The signal-to-noise ratio (SNR) is estimated at the receiver from five oscilloscope acquisitions after the coherent mixer, each containing 10^6 samples and stored for offline processing.

To decompose the noise contributions, the amplified spontaneous emission (ASE) noise-limited SNR (SNR_{ASE}) is measured with an optical spectrometer by replacing the CUT with a depolarized laser to access the ASE noise floor.

In addition, the transceiver (TRx) implementation penalty is characterized in a back-to-back configuration, yielding the TRx-limited SNR (SNR_{TRx}) and the corresponding gap-to-Shannon efficiency metric (η). Based on these measurements, the external SNR (SNR_{EXT}), representing the transmission-induced noise after removing TRx contributions, is obtained as:

$$\frac{1}{\text{SNR}_{\text{EXT}}} = \frac{\eta}{\text{SNR}_{\text{TOT}}} - \frac{1}{\text{SNR}_{\text{TRx}}}.$$

The final SNR_{EXT} is averaged over the two polarization tributaries. In the following, variance (VAR) is defined as the inverse of the SNR.

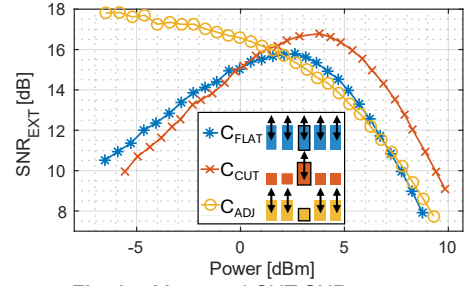


Fig. 3: Measured CUT SNR_{EXT} versus power for the three profiles used in experiments.

Experimental Data Acquisition

To obtain the ground-truth %SPM and %XPM values for experimental transmissions, we use the method described in [8], which enables the disaggregation of different propagation impairments. This procedure provides the target labels used for training the neural network.

We consider the transmission of WDM signals under different power profile configurations, designed to isolate SPM and XPM:

1) C_{FLAT} : flat profile with channels power ranging from -7 to 9 dBm; its nonlinear threshold (NLT) is denoted as $P(\text{NLT})$,

2) C_{CUT} : the CUT varies from -6 to 10 dBm while the 14 adjacent channels remain at $P(\text{NLT})$,

3) C_{ADJ} : the adjacent channels vary from -7 to 9 dBm while the CUT remains at $P(\text{NLT})$.

Fig. 3 shows the CUT's SNR_{EXT} , averaged over five acquisitions, versus launch power for the three configurations, for 32 GBd after 900km.

To separate the different noise contributions, the transmission external variance is modeled as:

$$\text{VAR}_{\text{EXT}} = \text{VAR}_{\text{ASE}} + \text{VAR}_{\text{NLI}} + \text{VAR}_{\text{RES}}, \text{ with } \text{VAR}_{\text{EXT}} = \frac{1}{\text{SNR}_{\text{EXT}}}.$$

VAR_{ASE} represents ASE noise, while VAR_{RES} accounts for power-independent residual impairments, such as the guided acoustic wave Brillouin scattering (GAWBS) penalty and the equalization-enhanced phase noise (EPPN). The nonlinear contributions are modeled as:

$\text{VAR}_{\text{SPM}} = \alpha_{\text{SPM}} * P_{\text{CUT}}^2$, $\text{VAR}_{\text{XPM}} = \beta_{\text{XPM}} * P_{\text{ADJ}}^2$, where P_{CUT} is the CUT power and P_{ADJ} is the averaged power of the 14 adjacent channels.

The nonlinear coefficients α_{SPM} and β_{XPM} are estimated by first subtracting ASE contribution from the total variance to obtain $\text{VAR}_{\text{EXT-ASE}}$, and then fitting a quadratic model as a function of power. This fit yields α_{SPM} , β_{XPM} , and VAR_{RES} and is repeated for each transmission configuration (different symbol rates and reaches).

Finally, the impairment ratios are given by:

$$\% \text{SPM} = 100 * \alpha_{\text{SPM}} * P_{\text{CUT}}^2 / \text{VAR}_{\text{EXT}},$$

$$\% \text{XPM} = 100 * \beta_{\text{XPM}} * P_{\text{ADJ}}^2 / \text{VAR}_{\text{EXT}},$$

$$\% \text{NLI} = \% \text{SPM} + \% \text{XPM}.$$

The distortion signal is obtained using a data-aided receiver by comparing the transmitted and

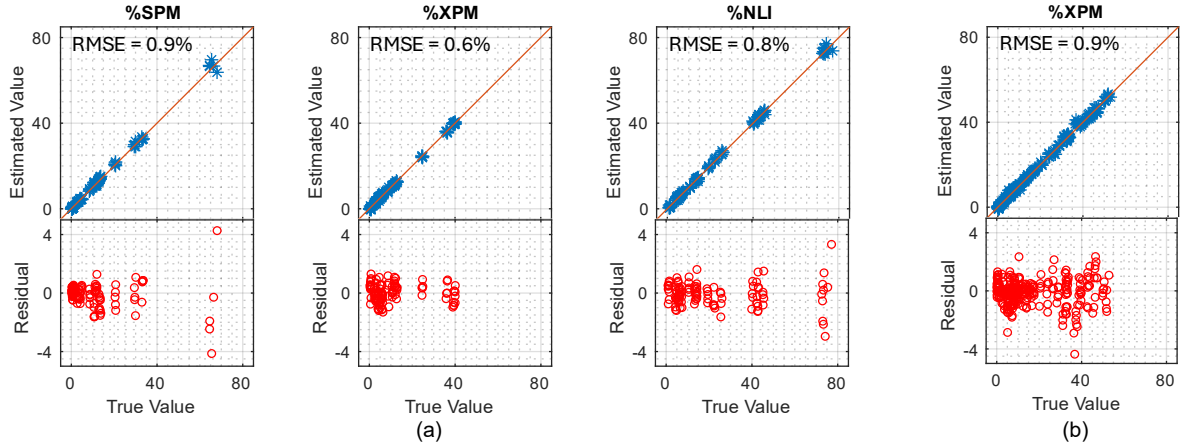


Fig. 4: Scatter plots and corresponding residual plots (shown below, same x-axis) for (a) the estimated %SPM, %XPM, and %NLI at 32 GBd after 3 loops, and (b) the estimated %XPM at 32, 39, and 46 GBd after 2 and 3 loops.

received symbols after the DSP chain. Alternatively, decision-directed detection can be used by replacing the known transmitted symbols with detected symbol decisions.

PSD_Q and PSD_I are then computed from the real and imaginary parts of the distortion signal using Welch's overlapped segment averaging estimator [9], with a 50-sample rectangular window.

To mitigate transceiver (TRx) noise contributions, the TRx-induced spectral component is assumed to be white and is subtracted from the measured spectra, yielding the corrected PSDs:

$$PSD_Q = \eta * PSD_Q^{withTRx} - 0.5 * PSD_{TRx},$$

$$PSD_I = \eta * PSD_I^{withTRx} - 0.5 * PSD_{TRx}.$$

Neural Network Results

The dataset was split into training, validation, and testing sets using a 70/15/15 ratio. Since five acquisitions were collected for each launch power profile, the testing set was constructed such that it contains launch powers not seen during training. We note that both training and test data originate from the same experimental setup.

In the first test scenario, the neural network was trained on 330 input–output pairs corresponding to 32 GBd transmissions over three loops (900 km) for the three power configurations (Fig. 3 legend). The model was evaluated using 5-fold cross-validation. Predictions for %SPM, %XPM, and %NLI from one fold are shown in Fig. 4 with corresponding residuals (estimated – true). The residuals are centered around zero, indicating no systematic bias, with a slight increase in error at higher %SPM values due to fewer training samples in this range. Across all folds, RMSE values are $1.1 \pm 0.2\%$, $0.6 \pm 0.1\%$, and $1.0 \pm 0.2\%$ for %SPM, %XPM, and %NLI.

In the second test scenario, the model is trained on transmissions over two and three loops (600 km and 900 km) with symbol rates of 32, 39, and 46 GBd, resulting in 1405 input–output pairs. It is tested on unseen launch power while keeping the same distances and symbol

rates. The model achieves RMSE values of $1.4 \pm 0.3\%$, $1.1 \pm 0.2\%$, and $1.5 \pm 0.3\%$ for %SPM, %XPM, and %NLI. The corresponding %XPM predictions from one fold are shown in Fig. 4 (b), with residuals centered around zero and no dependence on the true values.

Although RMSE is expressed in percentage points, it represents an absolute deviation from the true impairment ratios rather than a relative error. For example, an RMSE of 1% on %NLI corresponds to a 0.1 dB deviation in SNR_{NLI} around the optimal operating point at the NLT.

A comparison with state-of-the-art XPM estimation methods is reported in Table 1. Under comparable conditions (flat power profile and $VAR_{XPM} < 23\text{dBm}$), the proposed method achieves a mean absolute error of ~ 0.4 dB, compared with ~ 1 dB in [5] and 0.5 dB in [6] (simulation-based), and 1.5 dB in [7] at high launch powers ($> 4\text{dBm}$).

The model exhibits consistent performance across different symbol rates and link reaches, indicating robustness to parameter variations.

Tab. 1: XPM power estimation RMSE versus state of the art.

Method	Error	Conditions	Data type
[5]	$\sim 1\text{dB}$	$VAR_{XPM} > -23\text{dBm}$, flat profile	Experiment
[6]	$< 0.5\text{dB}$	Wider VAR_{XPM} range	Simulation
[7]	$< 1.5\text{dB}$	Launch power $> 4\text{dBm}$	Experiment
This work	$\sim 0.4\text{dB}$	$VAR_{XPM} > -23\text{dBm}$, flat profile	Experiment

Conclusions

We presented a neural network for estimating the relative contributions of SPM and XPM nonlinear impairments in fiber optic transmission. Once trained, the proposed method operates without affecting data transmission and without requiring external monitors or pilot symbols. Using signal PSDs as input features, the model achieves an RMSE of 1% on experimental transmissions across varying power profiles, symbol rates, and transmission distances.

References

- [1] I. Roberts and J. M. Kahn, "Measurement-Based Optimization of Channel Powers With Non-Gaussian Nonlinear Interference Noise," in *Journal of Lightwave Technology*, vol. 36, no. 13, pp. 2746-2756, 1 July 1, 2018, DOI: 10.1109/JLT.2018.2822719.
- [2] H. Wessing, P. Skoda, M. N. Petersen, A. Pilimon, P. Rydlichowski, G. Roberts, R. Smets, J. Radil, J. Vojtech, R. Lund, Z. Zhou, C. Tziouvaras, and K. Bozorgebrahimi, "Alien wavelengths in national research and education network infrastructures based on open line systems: Challenges and opportunities," in *Journal of Optical Communications and Networking*, vol. 11, no. 3, pp. 118-129, March 2019, DOI: 10.1364/JOCN.11.000118.
- [3] Y. Jiang, A. Nespola, S. Straullu, F. Forghieri, S. Picciaccia, A. Tanzi, M. R. Zefreh, G. Bosco, and P. Poggiolini, "Experimental Test of a Closed-Form EGN Model Over C+L Bands," in *Journal of Lightwave Technology*, vol. 43, no. 2, pp. 439-449, 15 Jan. 15, 2025, DOI: 10.1109/JLT.2024.3455752.
- [4] R. Wang, S. Bidkar, F. Meng, R. Nejabati and D. Simeonidou, "Load-aware nonlinearity estimation for elastic optical network resource optimization and management," in *Journal of Optical Communications and Networking*, vol. 11, no. 5, pp. 164-178, May 2019, DOI: 10.1364/JOCN.11.000164.
- [5] W. Wang, A. Yang, P. Guo, Y. Lu, and Y. Qiao, "Joint OSNR and Interchannel Nonlinearity Estimation Method Based on Fractional Fourier Transform," in *Journal of Lightwave Technology*, vol. 35, 4497-4506 (2017), DOI: 10.1109/JLT.2017.2744666
- [6] Y. Zhao, Z. Tao, S. Oda, Y. Aoki and T. Hoshida, "Pilot based cross phase modulation power estimation," *2017 Optical Fiber Communications Conference and Exhibition (OFC)*, Los Angeles, CA, USA, 2017, pp. 1-3, DOI: [10.1364/OFC.2017.W1G.2](https://doi.org/10.1364/OFC.2017.W1G.2)
- [7] J. Zhou, A. Yang, P. Guo, L. Zhuang, S. Guo, Y. Qiao, "Experimental estimation of inter-channel nonlinearity power by using differential pilot," in *Optical Fiber Technology*, vol. 58, 2020, 102256, ISSN 1068-5200, DOI: [10.1016/j.yofte.2020.102256](https://doi.org/10.1016/j.yofte.2020.102256)
- [8] J. Girard-Jollet, J. -C. Antona, A. C. Meseguer, F. Boitier, P. Ramantanis, and G. Rekaya, "Experimental Disaggregation of Propagation Effects in Optical Links," in *Optical Fiber Communication Conference (OFC)* 2024, Technical Digest Series (Optica Publishing Group, 2024), paper Th2A.23. [10.1364/OFC.2024.Th2A.23](https://doi.org/10.1364/OFC.2024.Th2A.23)
- [9] P. Welch, "The use of fast Fourier transform for the estimation of power spectra: A method based on time averaging over short, modified periodograms," in *IEEE Transactions on Audio and Electroacoustics*, vol. 15, no. 2, pp. 70-73, June 1967, DOI: 10.1109/TAU.1967.1161901